

1. Using the Binomial Distribution

An ensemble model is a model that uses multiple different models to build a better model. E.g., a random forest classifier uses multiple decision trees to “vote” for the best answer. For this exercise, assume that an ensemble model uses majority vote to predict a binary target and that each individual model is independent of every other model. Use Python code to answer the following questions.

- (a) If an ensemble model is composed of 15 individual models that each have 63% accuracy, what is the ensemble model accuracy?
- (b) What would the accuracy of the 15 individual models need to be in order for the ensemble model to have 95% accuracy? Find the smallest whole number percentage so this is satisfied.
- (c) If each individual model has 63% accuracy, how many individual models are needed for the ensemble model to have 95% accuracy? Find the smallest odd number of models (so there is a clear majority), so that this is satisfied.

Hint: The **binom** package from **scipy.stats** is useful for this problem, but make sure you are using the correct functions for your calculations.

2. Working with Random Data

Suppose that a random variable X can take on values 1, 2, 3, 4, or 5.

- (a) Make up your own probability mass function (pmf) for X . Make sure it satisfies the requirements of a pmf.
- (b) Build a function that will generate 50 random X -values from the pmf you created in part (a) and return the mean of these values.
- (c) Now, build a function that will run your function in part (b), 1,000 times and store the means in a list.
- (d) Create a histogram of the means in the list from part (c). Describe the shape of this histogram. What theorem tells us that it will look the way it does?
- (e) What is the mean and standard deviation of your list of means? What are their theoretical values?

Note: You can combine as much as you like of parts (b) and (c) into a single function. They are separate in the problem statement to make sure what you need to do is clear.